INSTRUCTIONS

AF3060 INVERTER DRIVES

INVERTER ASSEMBLY - 331X324EAG01, G04
G02, G05

OPERATION — TROUBLE SHOOTING — REPAIR

GEK-62504

DIAGRAMS

INVERTER CONV. ASM.	ELEM	36C764816EB	
	CONN	36D865531EA	(G01, G04)
		36D865520EB	(G02, G05)
PHASE MODULE	ELEM	36C764814EA	
	CONN	36C764815EA	
RECTIFIER MODULE ASM.	ELEM	36C764812AA	
	CONN	36C764813AA	
CURRENT FEEDBACK	ELEM	36C762857AB	
PULSE TRANSFORMER	ELEM	36C762856AB	
AUXILIARY SUPPRESSION	ELEM	36C762885AD	

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

GENERAL ELECTRIC

demark of the General Electric Co.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE ELEMENTARY DIAGRAM
AF – 3065 INVERTER ASM
FIRST MADE FOR STANDARD LINE
33IX324EAG01,602,604,605

6TB (PI) (AXI) (29) (30) (P3) (AX3) 6TB

THSW
PTB (PI)
PIR (PIR)
(SEE NOTE 8)
5TB (10) (11) THI TH2
SEE NOTE 7
RECT (RECTIFIER MODULE)
ELEM DIAG 36C764812AA
CONN DIAG 36C764813AA
CAT NO 33IX322AA 603
460VAC INPUT (L2) L2 (L3) L3
LI (SEE NOTE 2) (P3)
(P2) P2
CTF3
CTF
PIF R1 Q 2K 300W
CI CI
CPTB (PI) CPTB (P3)
CPL
APL BPL

ARP
R10 220K ACP1 225 MF
R11 220K ACP2 225 MF

(29) AUXILIARY SUPPRESSION ASSEMBLY ASA (30)
(AM) (BM) (CM) 3TB
(AN) (BN) (CN) 4TB

ACN1 225 MF R12 220K
ACN2 225 MF R13 220K
ARN

PMA (PHASE MODULE A)
ELEM DIAG 36C764814EA
CONN DIAG 36C764815EA
CAT NO. 33IX323EA602
PMB (SAME AS PHASE A)
PMC (SAME AS PHASE A)
PI P3 NO MO

CTCA CCA AN AL LCA
CTCB CCB BN BL LCB
CTCC CCC CN CL LCC
CTA CTB CTC
PTB (TI) TI
T2 (T2)
T3 (T3)

W R W R W R
5TB (1) (2) (3) (4) (5) (6)
W R W R W R

CURRENT FEEDBACK PANEL ASSEMBLY
ITB (PIF) ITB (QF) ITB (RFN) ITB (RFP) ITB (F3N)
R3 1K R2 1K R1 8H.2 ohms R15
ITB (PIL) ITB (QL) 47K R4 D2 D1 D3
ITB (PID) ITB (QD) ITB (RFD)
7 8 16 15

CURRENT FEEDBACK CARDS
ACF
BCF
CCF
K1 K2 K3 K4 J1 J2 J3 J4
17 18 19 20 21 22 23 24 25 26 27 28
SEE NOTE 6
.01MF 10K
SEE NOTE 16

THI TH2 (THI) (TH2)
IPL 2PL

SEE SHEET 2 FOR NOTES AND COMPONENT VALUES

FOLD
CONT ON SHEET
DRAWING NO
SH NO
REV NO. 0

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS	PRINTS TO:	
	58(3)23	JA(CD)
	50(BK)	IT(BW)
	5(QC)B	5AE(BW)
	5QC(BK)5	5L(BW)5
	5(TS)50	
	5(S)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY DEC.20,1974
APPROVALS Feb.24,1975
SPEED VARIATOR DIV OR DEPT ERIE, PA LOCATION
36C764816EB CONT ON SHEET 2 SH NO. 1

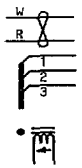
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE NOTES
AF-3065 INVERTER ASM
FIRST MADE FOR STANDARD LINE
331X324EAG01, 602, 604, 605

1. THE ASSEMBLY CATALOG NUMBER GROUP DISTINCTION, MODULE CATALOG NUMBERS AND SOME COMPONENT VALUES ARE DEPENDENT UPON KVA RATING. TABLE I LISTS VARIABLE COMPONENT VALUES.

2. THE L1 CHOKE IS MOUNTED REMOTE TO THE REMAINDER OF THE INVERTER ASSEMBLY. ITS VALUE IS DEPENDENT UPON KVA RATING AND IS LISTED IN TABLE I.

3. ELEMENTARY WIRING SYMBOLS



W / R — INDICATES TWISTED WIRES, COLOR INDICATED BY LETTER

1, 2, 3 — INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG

DOT INDICATES POSITIVE TERMINAL WHEN CURRENT FLOWS IN THE DIRECTION OF THE ARROW.

4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.

5. ON PRINTED CIRCUIT CARDS USED IN THIS ASSEMBLY, THE LETTERS "AA" AFTER THE BASIC CATALOG NUMBER INDICATES ORIGINAL DESIGN. SUBSEQUENT DESIGNS WITH THE SAME BASIC NUMBER AND GROUP NUMBER WITH THE SECOND LETTER CHANGED, SUCH AS "AB", "AC", "AD", ETC. ARE DIRECTLY INTERCHANGEABLE AND MAY BE SUPPLIED IN PLACE OF THE "AA" CARD. AN EARLIER DESIGN CANNOT REPLACE A LATER DESIGN, E.G. AN "AA" CANNOT REPLACE AN "AB".

6. THE NOTED RESISTOR VALUES ON THE CURRENT FEEDBACK CARD ARE DEPENDENT UPON KVA RATING. THE PARTICULAR CARD AND JUMPER CONNECTION IS FOUND IN TABLE I. THE CIRCUIT SHOWN IN DASHED LINES IS A REPRESENTATION OF THE ACTUAL CIRCUITRY AND IS IDENTICAL FOR ACF, BCF AND CCF.

7. RECTIFIER ASSEMBLY AND ASSOCIATED WIRE SHOWN IN DASHED LINES ARE OMITTED FOR G04 & G05.

8. **PIR JUMPER SHOWN AS A DASHED LINE IS REQUIRED FOR G01 AND G02.**

TABLE I

KVA RATING	L1	C1	C1 & C1	CCA, CCB, CCC	LCA, LCB, LCC	RF RF3	ACF BCF CCF	CURRENT FEEDBACK JUMPER KVA	CURRENT FEEDBACK JUMPER CONNECTION	ARP ARN
30	4MH	4250MF	2125MF	4MF	90MH	120Ω	193X 199BAG01	10	XA	2500Ω
								15	XB	
								30	XC	
60	3.2MH	8500MF	4250MF	8MF	40MH	56.2Ω	193X 199BAG02	30	XA	1000Ω
								40	XB	
								60	XC	

REV NO. C

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

2 CHG. RF3 FROM 150Ω TO 120Ω PER C. Gendreau JUNE 23, 1975

PRINTS TO

5B(6)M	5J(BW)
5R(2)BW	TS(SO)
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

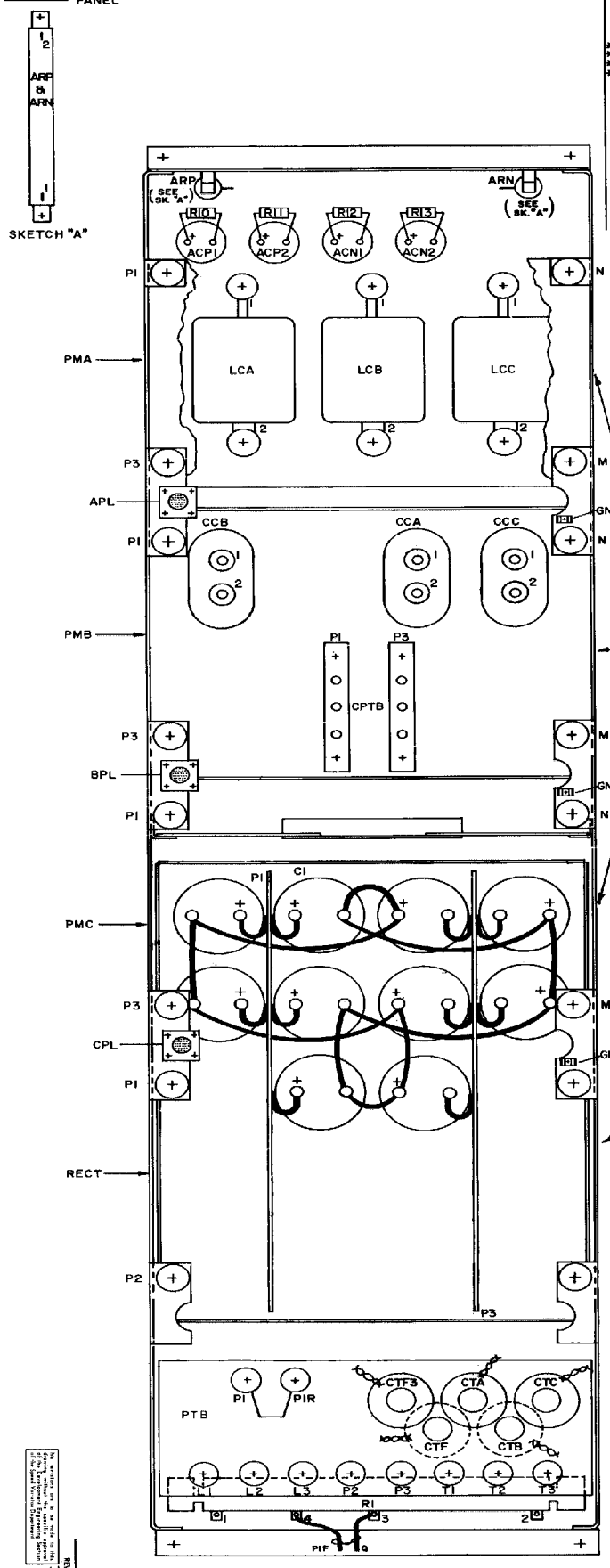
MADE BY G. Marsh DEC 26, 1974
MODIFIED G. Marsh FEB. 24 1975
APPROVALS

SPEED VARIATOR DIV OR DEPT ERIE, PA. LOCATION

36C764816EB CONT ON SHEET FL. SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FF-608-F (9-68) PRINTED IN U.S.A.



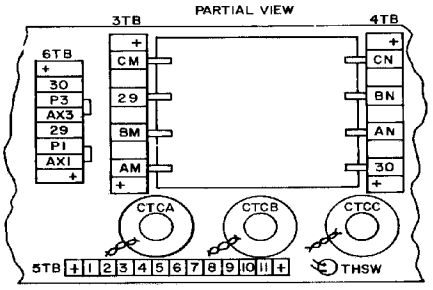
FRONT VIEW OF INVERTER ASM WITH RECTIFIER AND PHASE MODULES REMOVED.

WIRE TABLE	
WIRE	CONNECTION
	INPUT POWER WIRE
	#8 BLACK FLEX
* L1	PTB(L1),RECT(L1)
* L2	PTB(L2),RECT(L2)
* L3	PTB(L3),RECT(L3)
* P2	PTB(P2),RECT(P2)
	POWER WIRE
	#8 BLACK FLEX
P3-4	CPTB(P3),THRU CTF,CI(P3)
AL	CCA (2), 2TURN THRU CTCA BOT. TO TOP, LCA (2)
BL	CCB(2), 2 TURN THRU CTCB BOT.TO TOP, LCB(2)
CL	CCC(2), 2 TURN THRU CTCC BOT. TO TOP, LCC(2)
	L.H. TOP POWER WIRE
	#8 BLACK FLEX
P1-3	PMA(P1),CPTB(P1)
P1-4	PMB(P1),CPTB(P1)

WIRE TABLE CONT'D	
WIRE	CONNECTION
P3-1	PMA(P3),CPTB(P3)
	L.H. BOTTOM POWER WIRE
	#8 BLACK FLEX
P1-2	PTB(P1),CPTB(P1)
*P1-1	PTB(P1B),RECT(P1)
*P1B	PTB(P1B) PTB(P1)
	#8 BLACK FLEX
P1-6	PTB(P1),CI(P1)
P1-5	PMC(P1),CPTB(P1)
P3-2	PMB(P3),CPTB(P3)
P3-3	PMC(P3),CPTB(P3)
	R.H. POWER WIRE
	#8 BLACK FLEX
AN-1	PMA(N),CCA (1)
BN-1	PMB(N),CCB (1)
CN-1	PMC(N),CCC (1)
T1-1	LCA(1),PMA(M),THRU CTA,PTB(T1)
T2-1	LCB(1),PMB(M),THRU CTB,PTB(T2)

WIRE TABLE CONT'D	
WIRE	CONNECTION
T3-1	LCC(1),PMC(N),THRU CTC,PTB(T3)
P3-5	CPTB(P3) THRU CTF3,PTB(P3)
	ASA LEADS
BN-2	4TB(BN),2TURNS THRU CTCB (BOTTOM TO TOP),CCB(1)
AN-2	4TB(AM),2TURNS THRU CTCA (BOTTOM TO TOP),CCA(1)
CN-2	4TB(CM),2TURNS THRU CTCC (BOTTOM TO TOP),CCC(1)
	ASA CHOKE LEADS
AM-1	3TB(AM),LCA(1)
BM-1	3TB(BM),LCB(1)
CM-1	3TB(CM),LCC(1)
	CONTROL HARNESS
TH1	5TB(10),2TB(TH1)
TH2	5TB(11),2TB(TH2)
	#16 DELTABESTON
P1-9	CPTB(P1),R1(1)
P3-8	CPTB(P3),R1(2)
Q-1	C1(Q),R1(3)
	#18 RED & WHITE TWISTED
Q-2	R1(3),1TB(QF)
P1F	R1(F), 1TB(P1F)

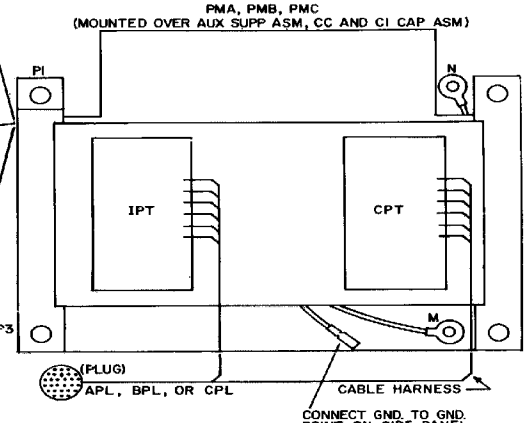
PARTIAL VIEW



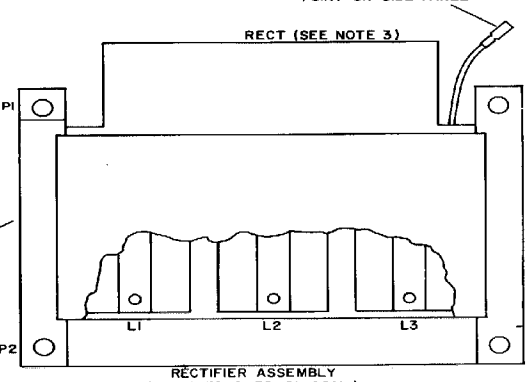
AUXILIARY SUPPRESSION ASSEMBLY (MOUNTED OVER LC CHOKE ASM.)

CURRENT TRANSFORMER LEADS	
LEAD	CONNECTION
CTA(WH)	ACF(J3)
CTA(RD)	ACF(J4)
CTB(WH)	BCF(J3)
CTB(RD)	BCF(J4)
CTC(WH)	CCF(J3)
CTC(RD)	CCF(J4)
CTF(WH)	1TB(RFP)
CTF(RD)	1TB(RFN)
CTF3(WH)	1TB(RFP)
CTF3(RD)	1TB(F3N)

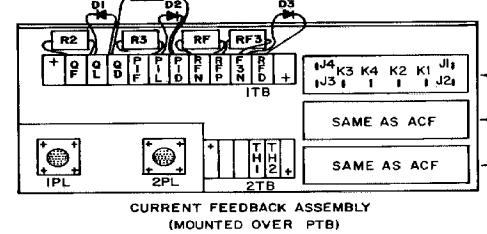
PMA, PMB, PMC
(MOUNTED OVER AUX SUPP ASM, CC AND CI CAP ASM)



FIRING SIGNAL CONTROL	
	MOUNT CONNECTORS APL, BPL, CPL, 1PL, 2PL, AND CONNECT WIRES:
QQ	1TB(QQ)
FID	1TB(PID)
RFD	1TB(RFD)
RFN	1TB(RFN)
ACF1P	ACF(K1)
ACF2N	ACF(K2)
ACF3P	ACF(K3)
ACF4N	ACF(K4)
BCF1P	BCF(K1)
BCF2N	BCF(K2)
BCF3P	BCF(K3)
BCF4N	BCF(K4)
CCF1P	CCF(K1)
CCF2N	CCF(K2)
CCF3P	CCF(K3)
CCF4N	CCF(K4)
CTCAP	5TB(1),ACF(J1)
CTCAN	5TB(2),ACF(J2)
CTCBP	5TB(3),BCF(J1)
CTCBN	5TB(4),BCF(J2)
CTCCP	5TB(5),CCF(J1)
CTCCN	5TB(6),CCF(J2)
	ASA TB LEADS
	#16 BLACK
P1-7	CPTB(P1)
P3-7	CPTB(P3)
P1-8	ACN1(+)
P3-8	ACP1(-)
29-2	ACP2(+)
30-2	ACN2(-)
	JUMPER CONN
ACN	ACN1(-),ACN2(+)
ACP	ACP1(-),ACP2(-)
29-1	6TB(29),3TB(29)
30-1	6TB(30),4TB(30)
	#16 DELTABESTON
29-3	ARP(1)
AX1	ARP(2)
30-3	ARN(1)
AX3	ARN(2)



RECTIFIER ASSEMBLY (MOUNTED OVER CI ASM.)



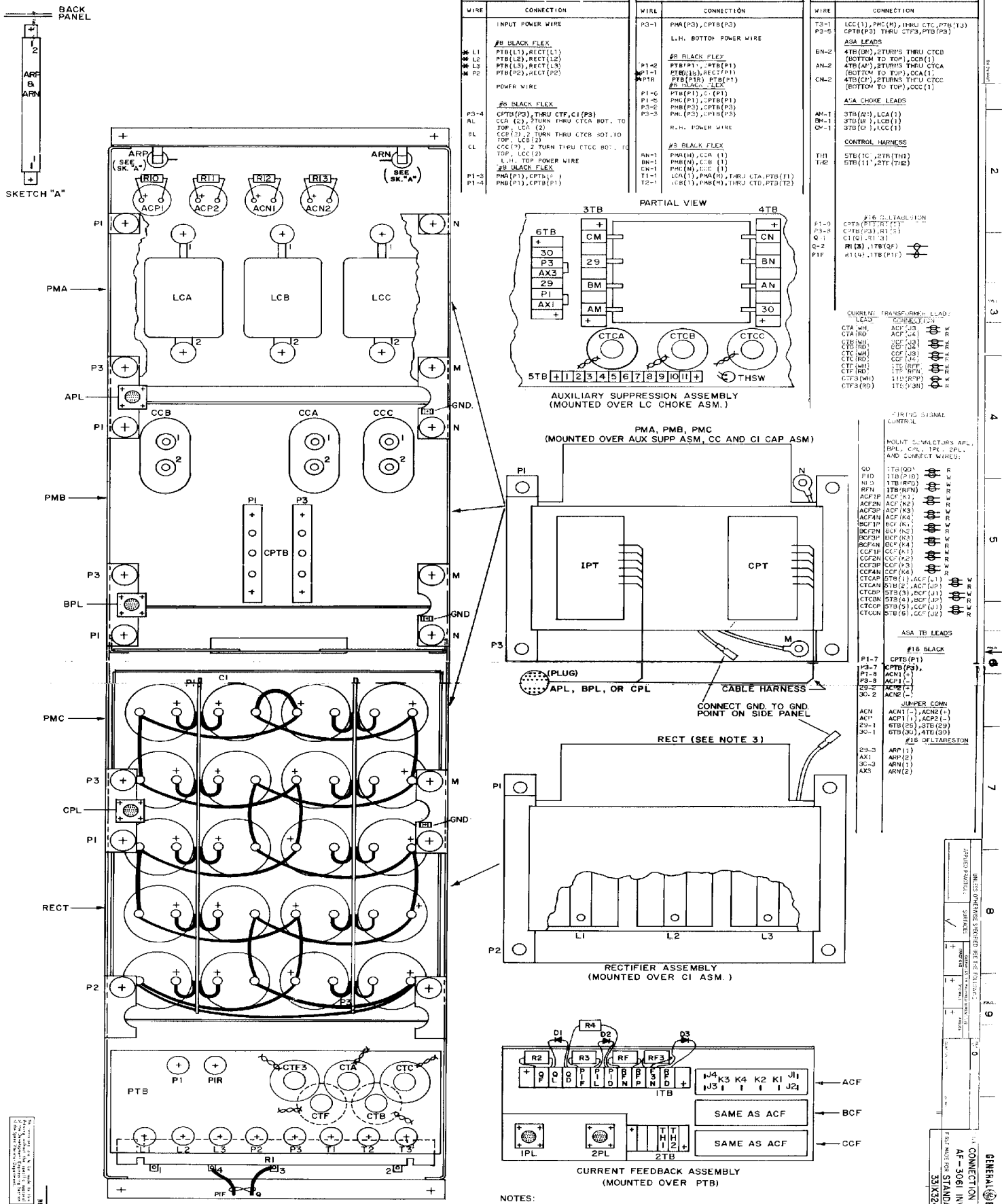
CURRENT FEEDBACK ASSEMBLY (MOUNTED OVER PTB)

NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE INVERTER ASSEMBLY TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN OMITTED FOR CLARITY.
2. PHASE MODULES AND RECTIFIER MODULE SLIDE ONTO THE STUDS IN THE INDICATED SPACES.
3. RECTIFIER ASSEMBLY AND ASSOCIATED WIRES MARKED WITH (*) ARE REMOVED FOR G04.
4. FAN ASSEMBLY SUPPLIED WHEN SPECIFIED. SEE SYSTEM DIAGRAMS FOR ACTUAL WIRING.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN DEC. 18, 1974
APPROVED Feb. 15, 1975
SPEED VARIATOR DEPT. ERIE, PA.
36D86553IE A
REVISIONS
PRINTS TO
GENERAL ELECTRIC
TITLE CONNECTION DIAGRAM AF-3063 INVERTER ASSEMBLY FIRST MADE FOR STANDARD LINE 33JX324EAG01&G04
360D86553IEA



WIRE TABLE	
WIRE	CONNECTION
	INPUT POWER WIRE
	#8 BLACK FLEX
* L1	PTB(L1),RECT(L1)
** L2	PTB(L2),RECT(L2)
* L3	PTB(L3),RECT(L3)
* P2	PTB(P2),RECT(P2)
	POWER WIRE
	#6 BLACK FLEX
P3-4	CPTB(P2),THRU CTF,CI(P3)
AL	CCA (2),2TURN THRU CTCA BOT. TO TOP, LCA (2)
BL	CCB(3),2 TURN THRU CTCB BOT.TO TOP, LCB(2)
CL	CCC(7), 2 TURN THRU CTCC BOT. TO TOP, LCC(2)
	L.H. TOP POWER WIRE
	#8 BLACK FLEX
P1-3	PMA(P1),CPTB(P1)
P1-4	PMB(P1),CPTB(P1)

WIRE TABLE CONT'D	
WIRE	CONNECTION
P3-1	PMA(P3),CPTB(P3)
	L.H. BOTTOM POWER WIRE
	#8 BLACK FLEX
P1-2	PTB(P1),CPTB(P1)
*P1-1	PTB(L1),RECT(P1)
*PTR	PTB(P1R) PTB(P1)
	#8 BLACK FLEX
P1-6	PTB(P1),CI(P1)
P1-5	PMC(P1),CPTB(P1)
P3-2	PMB(P3),CPTB(P3)
P3-3	PMC(P3),CPTB(P3)
	R.H. POWER WIRE
	#8 BLACK FLEX
AN-1	PMA(N),CCA (1)
BN-1	PMB(N),CCB (1)
CN-1	PMC(N),CCC (1)
T1-1	LCA(1),PMA(M),THRU CTA,PTB(T1)
T2-1	LCB(1),PMB(M),THRU CTB,PTB(T2)

WIRE TABLE CONT'D	
WIRE	CONNECTION
T3-1	LCC(1),PMC(M),THRU CTC,PTB(T3)
P3-5	CPTB(P3) THRU CTF3,PTB(P3)
	ASA LEADS
BN-2	4TB(DN),2TURNS THRU CTCB (BOTTOM TO TOP),CCB(1)
AN-2	4TB(AP),2TURNS THRU CTCA (BOTTOM TO TOP),CCA(1)
CN-2	4TB(CP),2TURNS THRU CTCC (BOTTOM TO TOP),CCC(1)
	ASA CHOKE LEADS
AM-1	3TB(AP),LCA(1)
BM-1	3TB(BP),LCB(1)
CM-1	3TB(CP),LCC(1)
	CONTROL HARNESS
TH1	5TB(1C ,2TB(TH1)
TH2	5TB(11",2TE(TH2)

#16 GUILTASTON	
P1-9	CPTB(P1),R1(5)
P3-8	CPTB(P3),R1(6)
Q-1	C1(G),R1(9)
Q-2	R1(8),1TB(QF)
P1F	R1(4),1TB(P1F)

CURRENT TRANSFORMER LEADS	
LEAD	CONNECTION
CTA (WH)	ACF(J3)
CTA (RD)	ACF(J4)
CTB (WH)	BCF(J3)
CTB (RD)	BCF(J4)
CTC (WH)	CCF(J3)
CTC (RD)	CCF(J4)
CTF (WH)	1TE(RFP)
CTF (RD)	1TE(RFN)
CTF3 (WH)	1TD(RFP)
CTF3 (RD)	1TE(P3N)

FIRING SIGNAL CONTROL	
	MOUNT CONNECTORS APL, BPL, CPL, 1PL, 2PL, AND CONNECT WIRES:
QD	1TB(QD)
P1D	1TB(P1D)
ND	1TB(RPD)
RFN	1TB(RFN)
ACF1P	ACF(K1)
ACF2N	ACF(K2)
ACF3P	ACF(K3)
ACF4N	ACF(K4)
BCF1P	BCF(K1)
BCF2N	BCF(K2)
BCF3P	BCF(K3)
BCF4N	BCF(K4)
CCF1P	CCF(K1)
CCF2N	CCF(K2)
CCF3P	CCF(K3)
CCF4N	CCF(K4)
CTCAP	STB(1),ACF(J1)
CTCAN	STB(2),ACF(J2)
CTCBP	STB(3),BCF(J1)
CTCBN	STB(4),BCF(J2)
CTCCP	STB(5),CCF(J1)
CTCCN	STB(6),CCF(J2)
	ASA TB LEADS
	#16 BLACK
P1-7	CPTB(P1)
P3-7	CPTB(P3),
P1-8	ACP1(+)
P3-8	ACP1(-)
29-2	ACN2(+)
30-2	ACN2(-)
	JUMPER CONN
ACN	ACN1(-),ACN2(+)
ACP	ACP1(+),ACP2(-)
29-1	6TB(29),3TB(29)
30-1	6TB(30),4TB(30)
	#16 DELTABESTON
29-3	ARP(1)
AX1	ARP(2)
30-3	ARN(1)
AX3	ARN(2)

NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE INVERTER ASSEMBLY TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN OMITTED FOR CLARITY.
2. PHASE MODULES AND RECTIFIER MODULE SLIDE ONTO THE STUDS IN THE INDICATED SPACES.
3. RECTIFIER ASSEMBLY AND ASSOCIATED WIRES MARKED WITH (*) ARE REMOVED FOR GOS.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN: C. Frank DEC 19, 1974
CHECKED: Feb 24 1975
ENGINEER: dab
SPEED VARIATOR
ERIE, PA.
36D865520EB

REVISIONS

PRINTS TO
5B1323 JA (CD)
5D(BK) (1TE(W))
5D(C)B BAE(BK)
5D(C)B BAE(BK)(WHS)
5(S)
5D(C)B BAE(BK)(WHS)
5(TS)(S0)

GENERAL ELECTRIC
CONNECTION DIAGRAM
AF-3061 INVERTER ASSEMBLY
STANDARD LINE
33X324EA602-G05
36D865520EB

GENERAL ELECTRIC

36C764814EA

CONT ON SHEET — SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

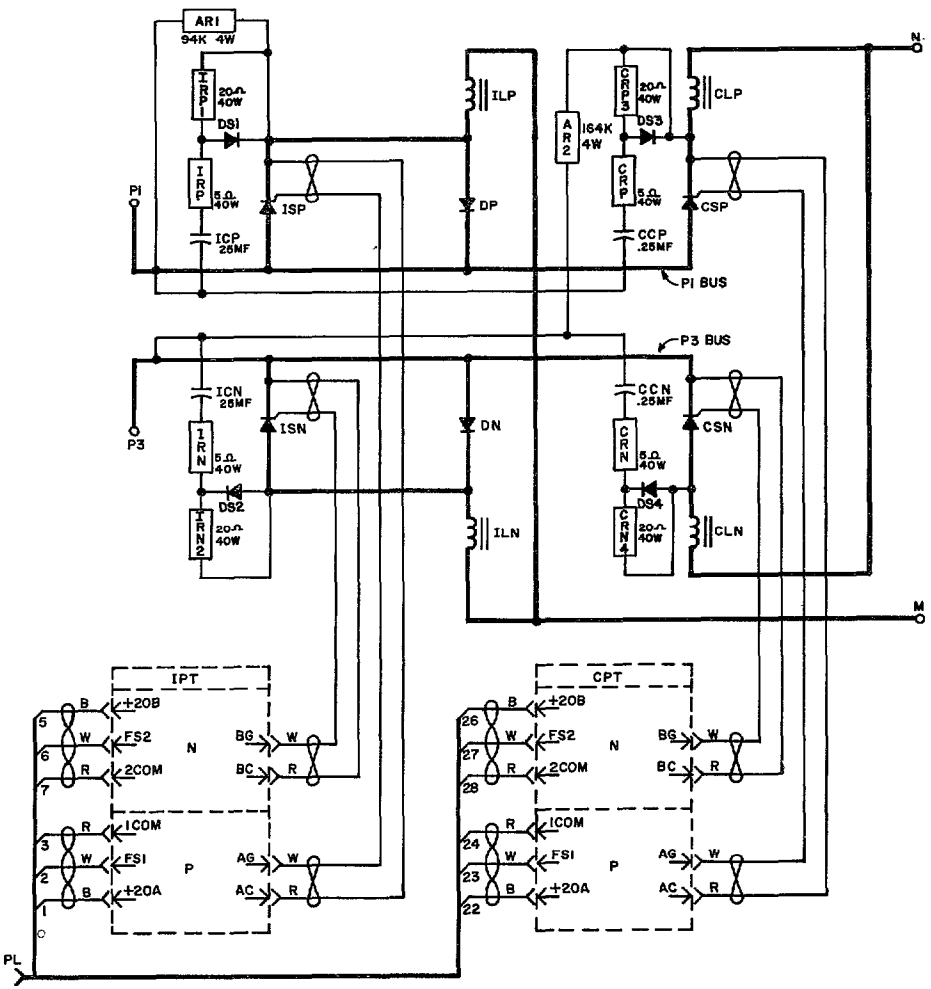
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM

AF-3063 PHASE MODULE

FIRST MADE FOR STANDARD LINE

331X323EAG02



NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES OF I OR C TO INDICATE INVERTER OR COMMUTATION SECTION. SUFFIXES OF P OR N INDICATE POSITIVE OR NEGATIVE. THE BASIC NOMENCLATURE IS:
   - C — CAPACITOR
   - D — DIODE
   - L — CHOKE
   - PT — PULSE TRANSFORMER PRINTED CIRCUIT CARD
   - PL — CONNECTOR PLUG
   - R — RESISTOR
   - S — SCR
   - DS — SNUBBER DIODE
2. PULSE TRANSFORMER CARD CIRCUITRY IS NOT SHOWN FOR SIMPLIFICATION FOR DETAILS REFER TO THE INSTRUCTION BOOK.
3. ELEMENTARY WIRING SYMBOLS



— INDICATES TWISTED WIRES, COLORS INDICATED BY LETTER



— INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG PL.

— QUICK DISCONNECT "STAB-ON" CONNECTION

4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTION OF THIS ASSEMBLY
5. THE ASSEMBLY CAT NO. GROUP DISTINCTION IS FOR MAXIMUM RATING

G02 — 60 KVA

REVISIONS

REV NO	DESCRIPTION	DATE	APPROVED
1		DEC 17,'76	SV-BHP

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

58(3)23	JA(CD)
5D(BK)	
5(QC)B	
5QC(BK)5	
5(TS)50	
5(S)	

MADE BY R.T. Mozak DEC. 20, 1974

ISSUED R. Mozak FEB 24, 1975

APPROVALS

SPEED VARIATOR — DIV OR DEPT

ERIE, PA. — LOCATION

36C764814EA — CONT ON SHEET — SH NO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

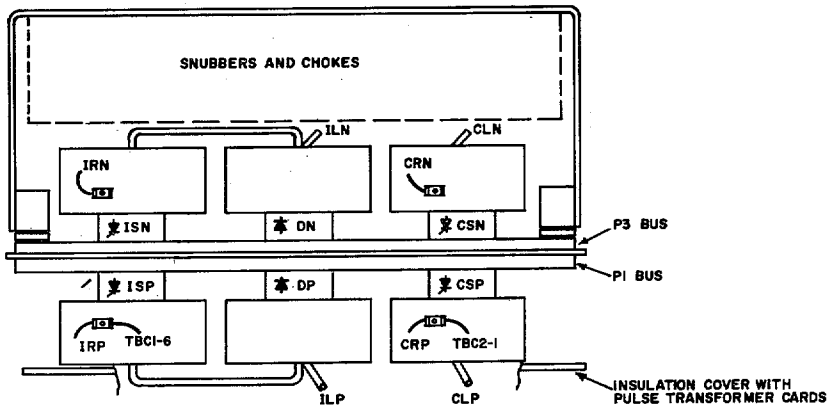
FF-809 P (4 72)
PRINTED IN U.S.A.

GENERAL ELECTRIC

36C764815EA
CONT ON SHEET — SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE CONNECTION DIAGRAM
AF-3065 PHASE MODULE
FIRST MADE FOR STANDARD LINE
331X323EA602

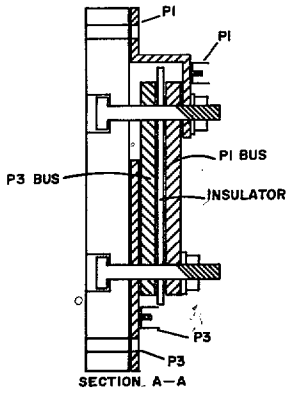


TOP VIEW OF PHASE MODULE

NOTES:

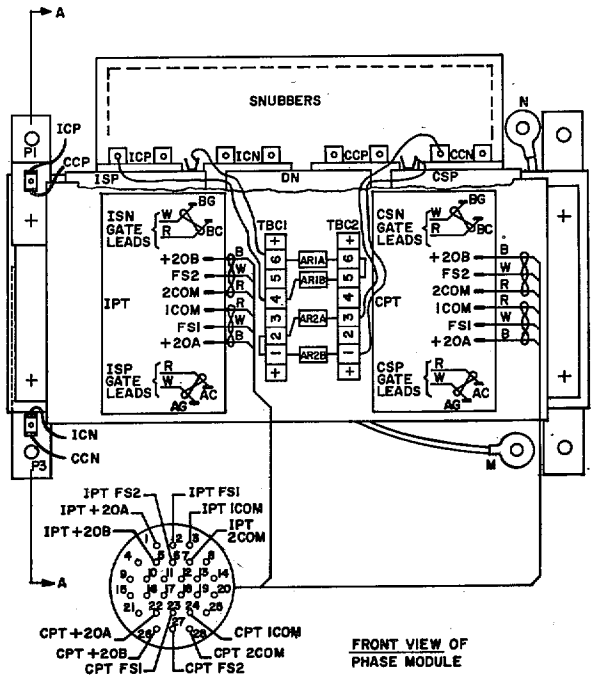
1. THIS DIAGRAM IS A REPRESENTATION OF THE PHASE MODULE TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN REMOVED FOR CLARITY.
2. GATE LEAD AND THE PULSE TRANSFORMER INPUT HARNESS ARE TWISTED WIRES INDICATED BY THE SYMBOL : W R B

   WIRE COLOR IS INDICATED FOR EASY IDENTIFICATION. GATE LEADS ARE WHITE, CATHODE LEADS ARE RED.



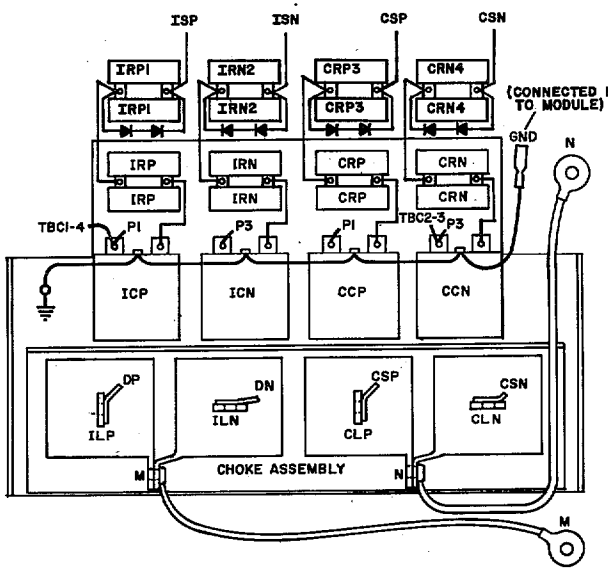
SECTION A—A

VIEW OF BUS ASSEMBLY



PL
(VIEWED FROM WIRE SIDE)

FRONT VIEW OF PHASE MODULE



FRONT VIEW OF SNUBBERS, CHOKES WITH HEATSINK ASSEMBLY REMOVED

REVISIONS

REV NO.	
Ø	
1	

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1 10-14-74 ADDED NOTE RRK SY-8745

PRINTS TO

5B(3)23	JA(CD)
5D(BK)	
5(QC)B	
5QC(BK)5	
5(TS)50	
5(S)	

MADE BY DEC. 20, 1974
ISSUED FEB 24, 1975
APPROVALS

SPEED VARIATOR — DIV OR DEPT
ERIE, PA — LOCATION
36C764815EA — CONT ON SHEET — SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

PF-800-F (4-72)
PRINTED IN U.S.A.

GENERAL ELECTRIC

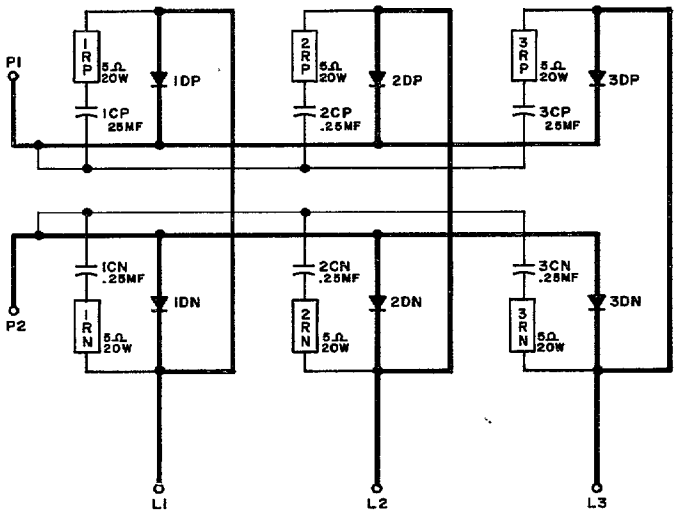
36C764812AA
CONT ON SHEET SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
AF-3061 RECTIFIER MODULE

FIRST MADE FOR STANDARD LINE
331X322AAG02, G03



NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES INDICATING PHASE (1, 2, OR 3) AND SUFFIXES OF P OR N INDICATING POSITIVE OR NEGATIVE. THE BASIC NOMENCLATURE IS:
   C – CAPACITOR
   D – DIODE
   R – RESISTOR
2. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
3. THE ASSEMBLY CAT NO. GROUP DISTINCTION IS FOR MAXIMUM RATINGS:

G02	50 KVA	230AC
G03	60 KVA	460AC

FOLD

36C764812AA
CONT ON SHEET SH NO

REV NO
0

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

5B(3)23	JA(CD)
5D(BK)	
5(QC)B	
5QCBK5	
5(TS)50	
5(S)	

MADE BY S. Mattia SEPT 27, 1972
ISSUED Nov 24 1971.

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION

36C764812AA
CONT ON SHEET SH NO

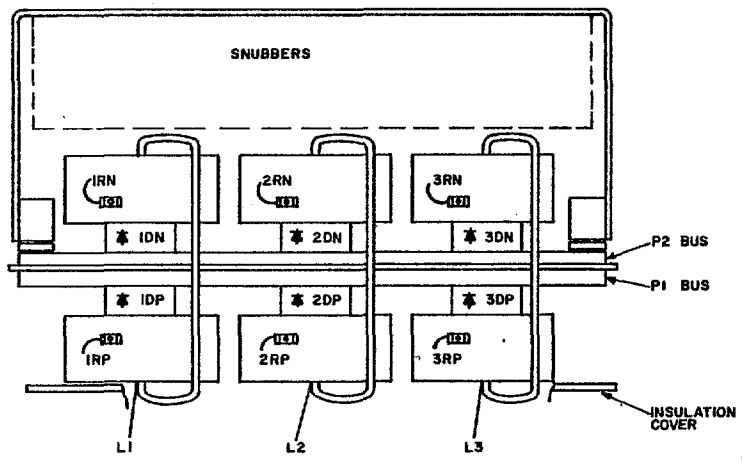
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

CONNECTION DIAGRAM
AF-3061 RECTIFIER MODULE

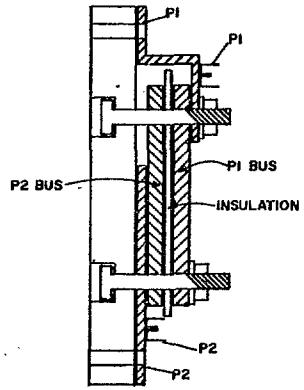
FIRST MADE FOR STANDARD LINE
331X322AAG02,003



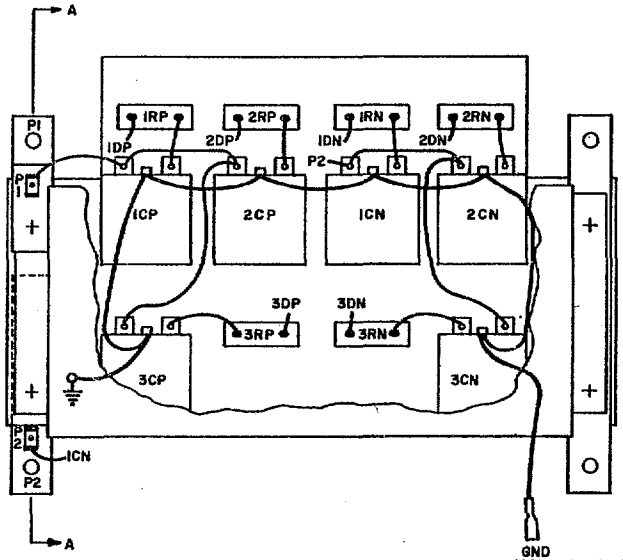
TOP VIEW OF RECTIFIER MODULE

NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE RECTIFIER MODULE TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN REMOVED FOR CLARITY.



SECTION A–A
VIEW OF BUS ASSEMBLY



FRONT VIEW OF RECTIFIER MODULE
WITH CUTAWAY SHOWING SNUBBERS

GND
(CONNECTED EXTERNAL
TO MODULE)

DRAWING NO. 36C764 813 AA
CONT ON SHEET
SH NO.
REV NC
1

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1 Per C.P. Mod Feb. 24, 1975 MB GP

PRINTS TO

5B(3)23	JA(CD)
5D(BK)	
5(QC)B	
5QCBK)5	
5(TS)50	
5(S)	

DRAWN BY J. Matter OCT. 10, 1972
ISSUED J.J. Kanne Nov 24, 1972
APPROVALS GPS KRP

SPEED VARIATOR DIV. OR DEPT
ERIE, PA. LOCATION

36C764 813 AA
CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

PRINTED IN U.S.A.

GENERAL ELECTRIC

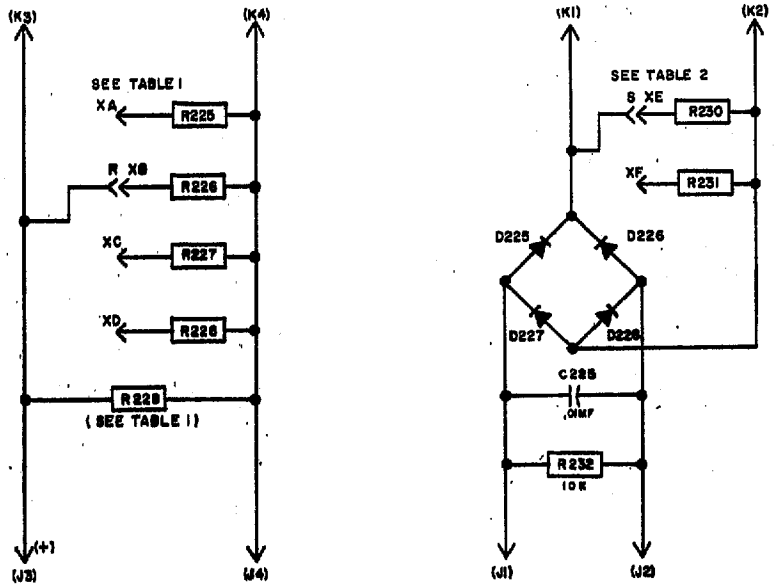
36C762857AB

SCHEMATIC DIAGRAM

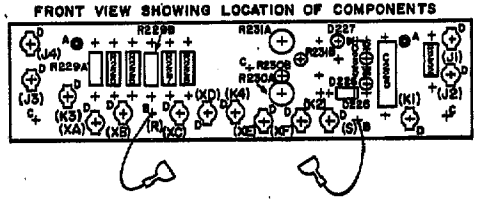
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
CURRENT FEEDBACK

FIRST MADE FOR STANDARD LINE
195X187AB601, 602, 603, 604



FRONT VIEW SHOWING LOCATION OF COMPONENTS



HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.062	2
C	.187	3
D	.101	14

NOTE: 1. CARD SIZE, 4.900 $^{+.000}_{-.018}$ X 1.300 $^{+.000}_{-.018}$

NOTES

TABLE 1

GROUP NUMBER	RESISTOR NOMENCLATURE AND VALUE				
	R225(XA)	R226(XB)	R227(XC)	R228(XD)	R229
G01	470Ω	56Ω	33.2Ω	22.1Ω	23.8Ω
G02	1K	39Ω	27.4Ω	18.2Ω	11.7Ω
G03	68Ω	33Ω	15Ω	10Ω	8.91Ω
G04	10K	39Ω	15Ω	10Ω	5.48Ω

TABLE 2

GROUP NUMBER	RESISTOR NOMENCLATURE AND VALUE	
	R230(XE)	R231(XF)
G01	63.6Ω	70.6Ω
G02	36.0Ω	49.1Ω
G03	27.9Ω	35.0Ω
G04	21.5Ω	28.2Ω

TABLE 3

GROUP NUMBER	RESISTANCE BETWEEN K3 & K4 USING JUMPER			
	XA	XB	XC	XD
G01	22.70Ω	16.7Ω	13.9Ω	11.5Ω
G02	11.5Ω	8.98Ω	8.19Ω	7.00Ω
G03	7.89Ω	7.05Ω	5.60Ω	4.71Ω
G04	5.47Ω	4.81Ω	4.01Ω	3.54Ω

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT	
		FRONT	BACK
FULL	9428519AB 004	—	—

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

5(C)(K)	5B(C)(+)
5B(2)22	5D(BK)
5(S)	5P(MF)
	5B(C)(-)
	5B(S)(C)
	5B(C)(R)

MADE BY W. Turner Apr 18 1972
ISSUED [illegible] Apr 18 1972
APPROVED 191 KDR

SPEED VARIATOR — DIV OR DEPT
ERIE — LOCATION

36C762857AB — CONT ON SHEET

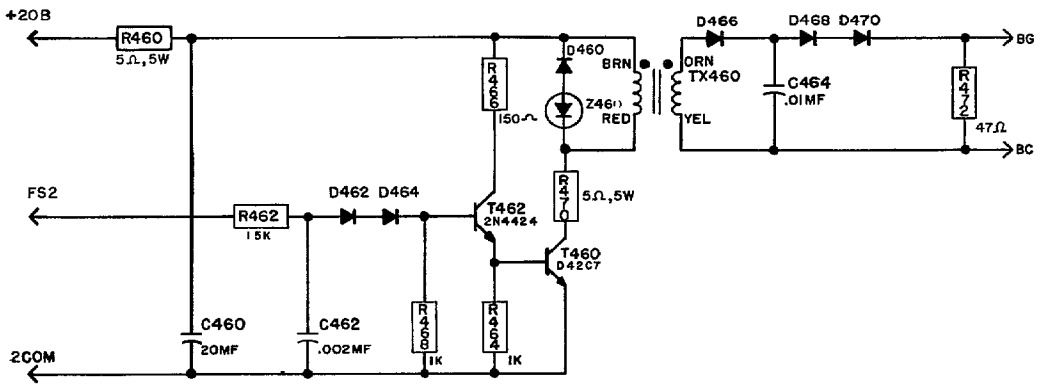
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

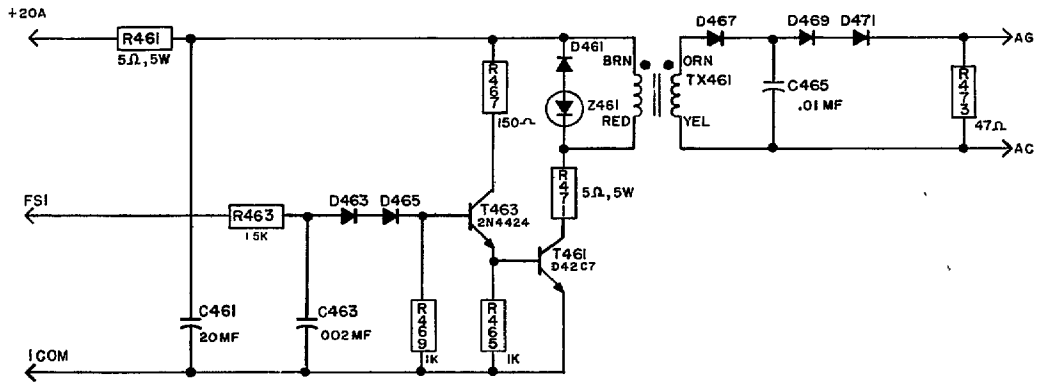
36C762857AB — CONT ON SHEET — REV NO. 0

GENERAL ELECTRIC 36C762856AB

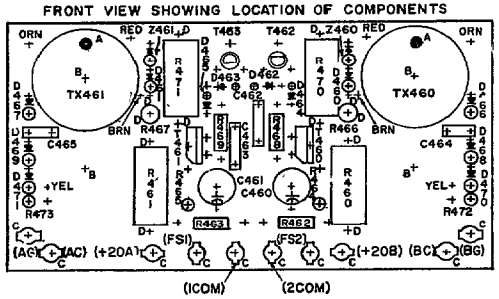
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -
APPLIED PRACTICES 718A515 | SUPERSEDES ✓ | FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE PRINTED CIRCUIT DIAGRAM
PULSE TRANSFORMER
FOR STANDARD LINE
193X186AB601





FRONT VIEW SHOWING LOCATION OF COMPONENTS



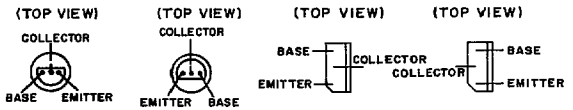
HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC	DIA.	QUAN
A	.157	2
B	.187	4
C	.101	10
D	.052	12

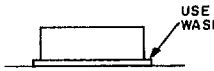
NOTE. 1 CARD SIZE $5.130 \pm \frac{.000}{.015}$ X $2.850 \pm .020$

TRANSISTOR LEAD SKETCHES





SKETCH FOR R460, R461, R470, R471



SKETCH FOR TX460, TX461

SCALE	ETCHED CIRCUIT BOARD DWG	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	942B521AA004	—	—

REV

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

5QC(BK)	5 (S)
	5D(BK)
	5B(2)
	5(TS)50
	5(QC)B

36C762856AB 0

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN E Novak Jan-25,1972
CHECKED D Juanos Jan 26,1972
ASSEMBLED
APPROVED

SPEED VARIATOR
ERIE, PA

36C762856AB

GENERAL ELECTRIC

36C762885AD

CONT ON SHEET | SH NO

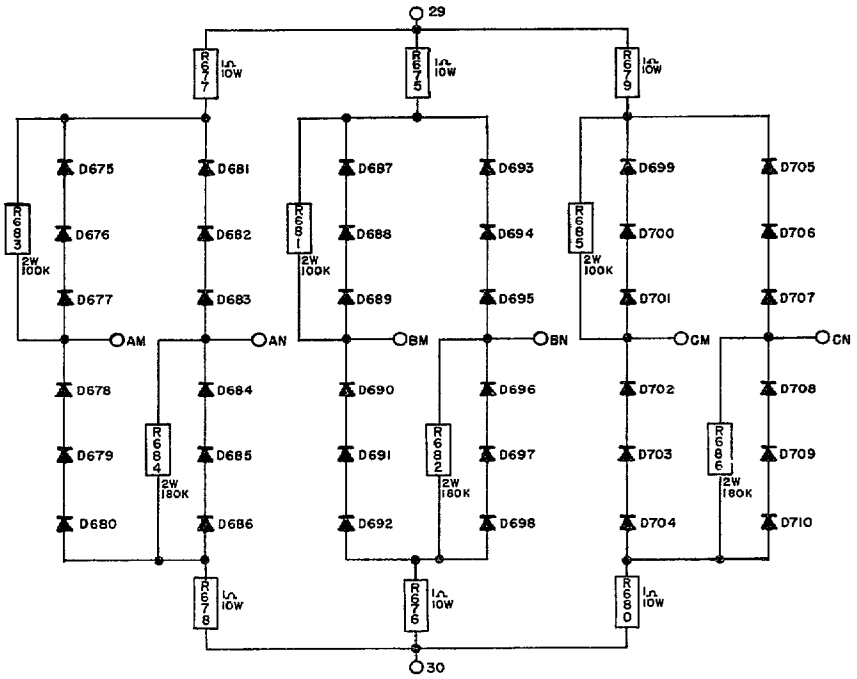
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

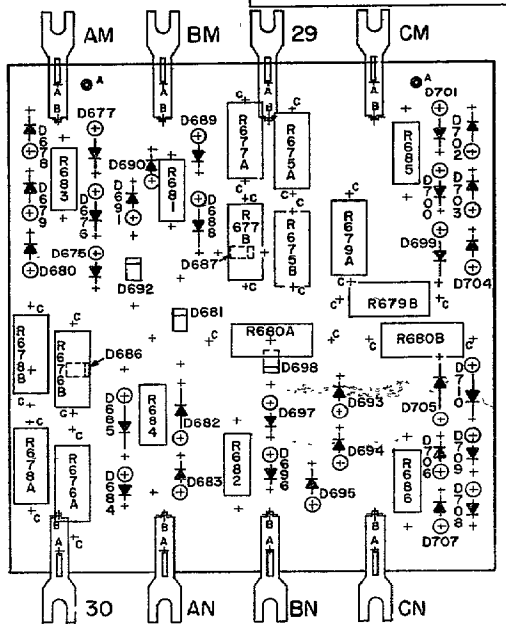
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
AUXILIARY SUPPRESSION

FIRST MADE FOR STANDARD LINE
193X193AD601

SCHEMATIC DIAGRAM





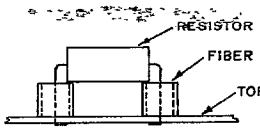
FRONT VIEW SHOWING LOCATION OF COMPONENTS

NOTE
1 CARD SIZE, $5.700^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

HOLE TABULATION

ALL HOLES .062 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC	DIA	QUAN
A	.157	10
B	101	8
C	052	24



MOUNTING SKETCH FOR RESISTORS
R675A, R675B, R676A, R676B, R677A, R677B, R678A, R678B, R679A, R679B, R680A, R680B, R681, R682, R683, R684, R685, R686

REV NO: 0

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT: FRONT	BACK
FULL	9428604AD004	—	—

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO	
SOC(BK)2	5 (S)
IT(BW)	5D(BK)
JA(CD)	5B(2)22
DS(REP)	5L(BW)5
	6(TS)50
	5(QC)8

MADE BY R. Mattis DEC 18, 1974
ISSUED Paul J. Nora Jr. MAR. 10, 1975
APPROVALS

SPEED VARIATOR
ERIE, PA
LOCATION

DIV OR DEPT
CONT ON SHEET
36C762885AD
SH NO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY